

CLASSWORK 7

Simple Linear Regression in One Variable

CPE 486/586

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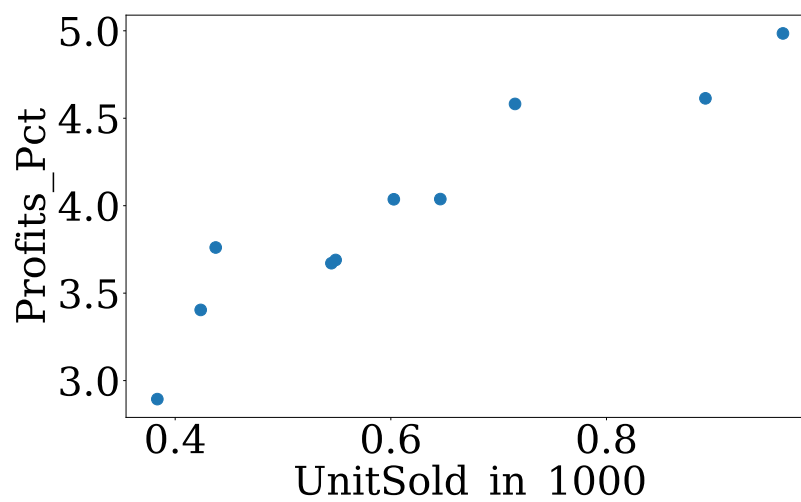
50 points

1 Simple Linear Regression

Consider the dataset that exhibits linear relationship between Units Sold in 1000 (predictor variable x), and profit in percentage (y):

UnitSold_in_1000 x_i	Profits_Pct y_i
0.54	3.69
0.72	4.58
0.6	4.03
0.54	3.67
0.42	3.40
0.64	4.03
0.43	3.76
0.89	4.61
0.96	4.98
0.38	2.89

$n = 10$.



Calculate the following quantity ($\sum_{i=1}^{10}$ is shorthand as Σ) (10 points):

1.

$$\sum x_i = \boxed{}$$

5.

$$\sum (x_i - \bar{x}) = \boxed{}$$

9.

$$\sum x_i^2 = \boxed{}$$

2.

$$\sum y_i = \boxed{}$$

6.

$$\sum (y_i - \bar{y}) = \boxed{}$$

10.

$$\sum y_i^2 = \boxed{}$$

3.

$$\bar{x} = \boxed{}$$

7.

$$\sum (x_i - \bar{x})^2 = \boxed{}$$

11.

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = \boxed{}$$

4.

$$\bar{y} = \boxed{}$$

8.

$$\sum (x_i y_i) = \boxed{}$$

In an ideal scenario, the data was obtained from a linear model $Y = Xw_1 + w_0$. The given data comes follows the $y_i = x_i w_1 + w_0 + \epsilon_i$ where ϵ_i is the error introduced by data gathering process such that $\epsilon_i \sim \mathcal{N}(\mu = 0, \sigma = 0.3)$.

We will fit our data into the linear model $y_i = x_i w_1 + w_0$ with the hope of minimizing the cost function (or error):

$$Q = \frac{1}{n} \sum_i (y_i - \hat{y})^2 \tag{1}$$

where $\hat{y} = x_i \hat{w}_1 + \hat{w}_0$.

The least-square solution gives the normal equation by solving:

$$\begin{aligned} \sum y_i &= n\hat{w}_0 + \hat{w}_1 \sum x_i \\ \sum x_i y_i &= \hat{w}_0 \sum x_i + \hat{w}_1 \sum x_i^2 \end{aligned} \tag{2}$$

$$\frac{\partial Q}{\partial w_0} = \quad (7)$$

(5 Points)

Setting above to 0 gives the following set of equations:

$$w_1 = \frac{n(\sum_{i=1}^n x_i y_i) - (\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n(\sum_{i=1}^n x_i^2) - (\sum_{i=1}^n x_i)^2} \quad (8)$$

$$w_0 = \frac{1}{n}(\sum_{i=1}^n y_i) - w_1 \frac{1}{n}(\sum_{i=1}^n x_i)$$

Putting the value in the above equations should give an estimated $\hat{w}_0$ and $\hat{w}_1$. Write down what you got in that Equation.

$$\hat{w}_1 = \quad (9)$$

$$\hat{w}_0 = \quad (10)$$

(5 Points)

Using $\hat{w}_0$ and $\hat{w}_1$, we can obtain $\hat{y}_i$, the estimated response. Using estimated response and actual response, we can calculate Sum of Square Error (SSE) and Total Square Sum (SSTO) as:

$$\begin{aligned} \text{SSE} &= \sum (y_i - \hat{y}_i)^2 = \\ \text{MSE} &= \frac{\text{SSE}}{n - 2} = \\ \text{SSTO} &= \sum (y_i - \bar{y})^2 = \end{aligned} \quad (11)$$

(2 Points)

Calculate the goodness of fit using the coefficient of determination R^2 :

$$R^2 = 1 - \frac{\text{SSE}}{\text{SSTO}} = \quad (12)$$

(2 Points)

Based on the value of R^2 how good fit is the linear model on the given dataset?

(2 Points)

Next, we calculate the standard deviation on the estimated coefficients or weight ($\hat{w}_1$, and $\hat{w}_0$).

$$s^2[\hat{w}_1] = \frac{\text{MSE}}{\sum (x_i - \bar{x})^2} = \quad (13)$$

(2 Points)

Now, we need to calculate the 95% confidence interval on the estimated $\hat{w}_1$ and $\hat{w}_0$ for which $\alpha = 0.05$ (use $t(0.975, 8) = 2.306$):

Now, consider a new data point $x_{new} = 0.47$ Units Sold (in 1000). Based on the linear regression model that we developed, what is the estimated/predicted profit percent for the given quantity

(5 Points)?

$$s^2[\hat{w}_0] = \text{MSE}\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2}\right) = \quad (14)$$

(2 Points)

$$\hat{w}_1 \pm t(1 - 0.05/2, 10 - 2)s[\hat{w}_1] \quad (15)$$

(2 Points)

$$\hat{w}_0 \pm t(1 - 0.05/2, 10 - 2)s[\hat{w}_0] \quad (16)$$

(2 Points)